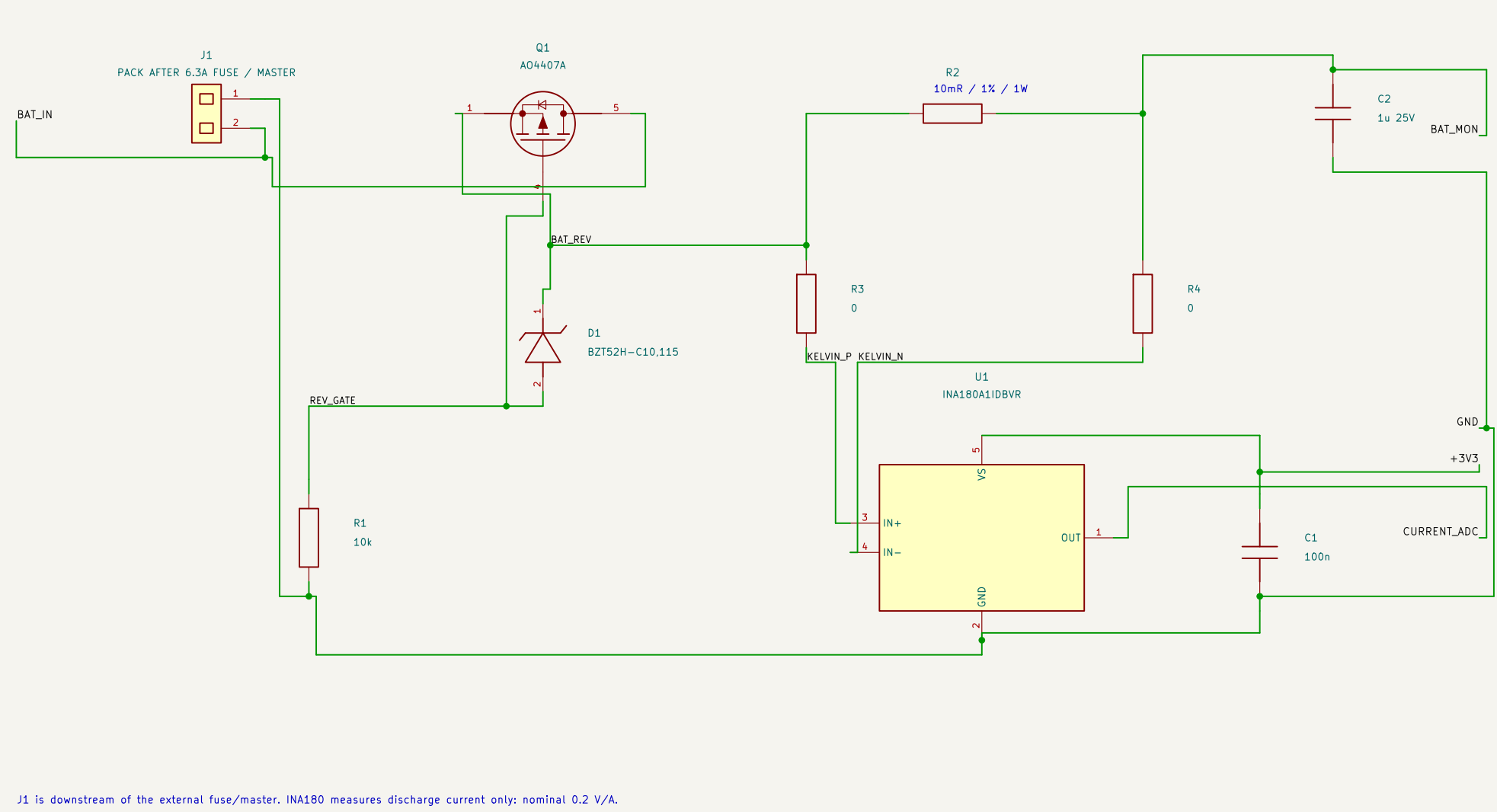


MORI POWER CONDITIONER | FUNCTIONAL CIRCUIT

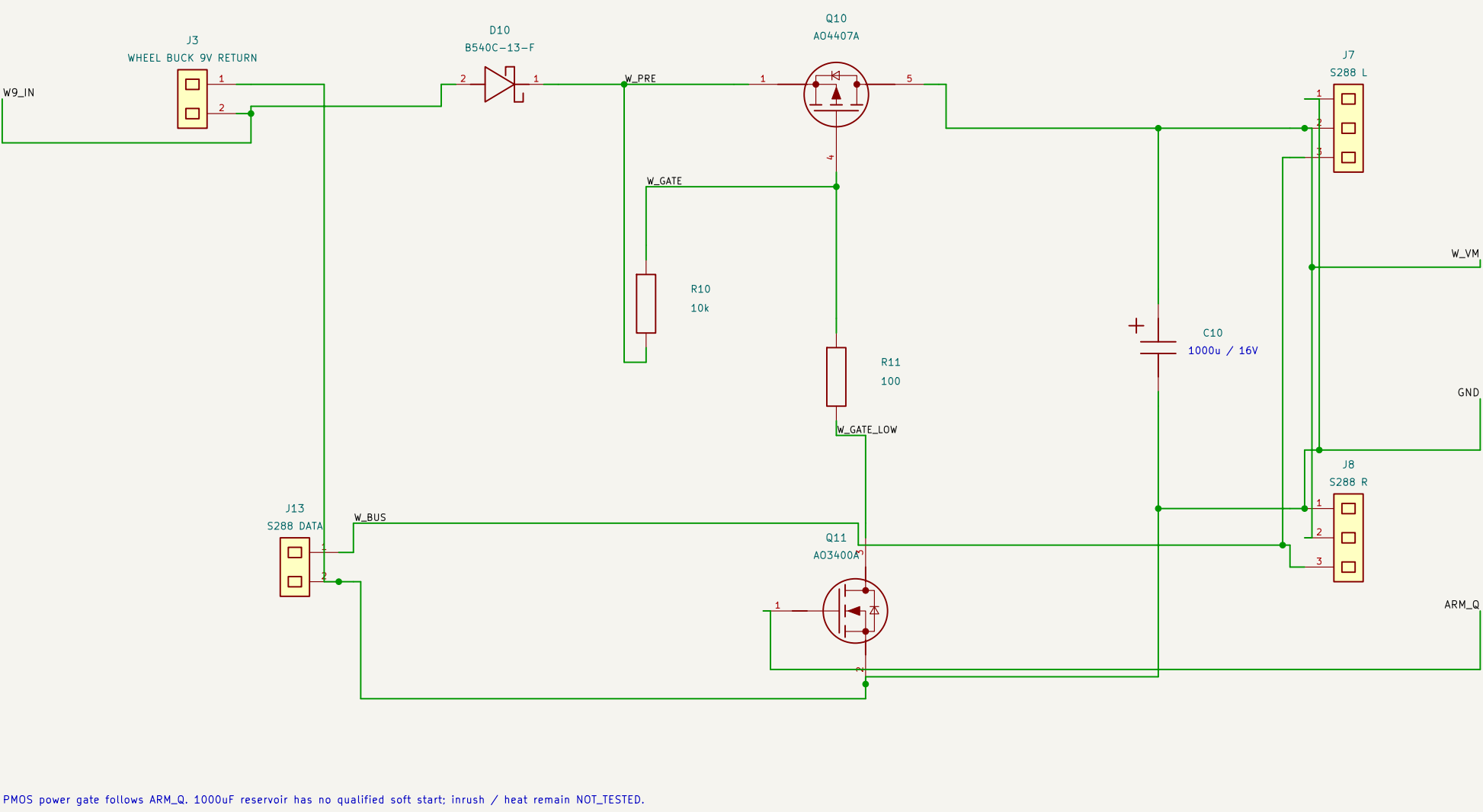
P1–S1 drawing revision, Protected 3S pack and buck converters are external. This board does not implement USB–C charging.

01 Pack input / reverse protection / discharge current



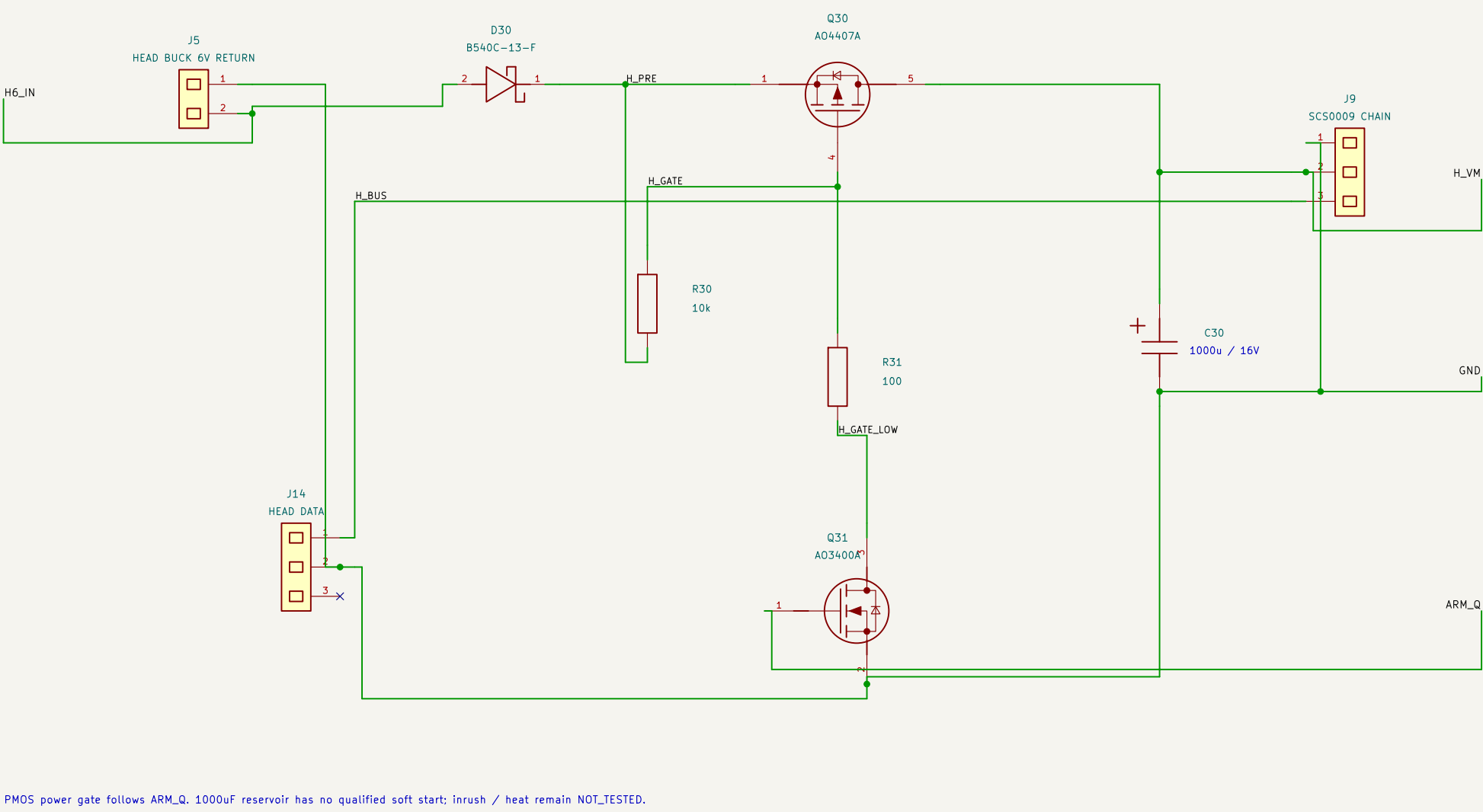
J1 is downstream of the external fuse/master. INA180 measures discharge current only; nominal 0.2 V/A.

02 Wheel rail / reverse blocking / hardware power enable



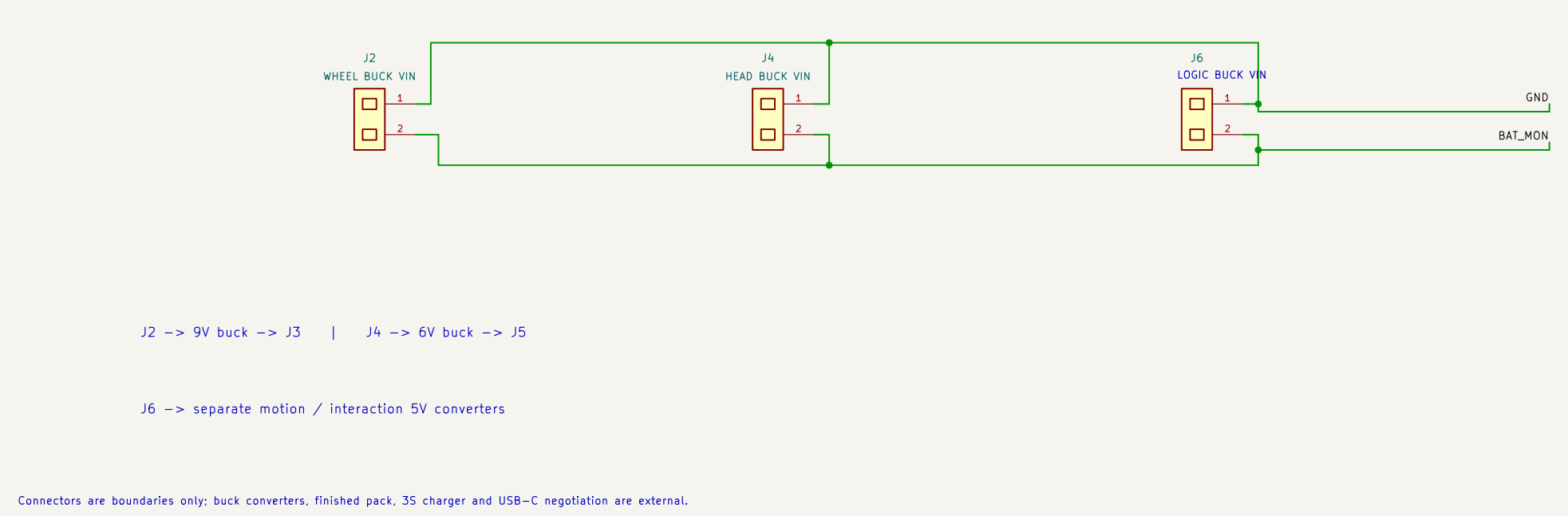
PMOS power gate follows ARM_Q. 1000uF reservoir has no qualified soft start; inrush / heat remain NOT_TESTED.

03 Head rail / reverse blocking / hardware power enable



PMOS power gate follows ARM_Q. 1000uF reservoir has no qualified soft start; inrush / heat remain NOT_TESTED.

04 External buck input distribution

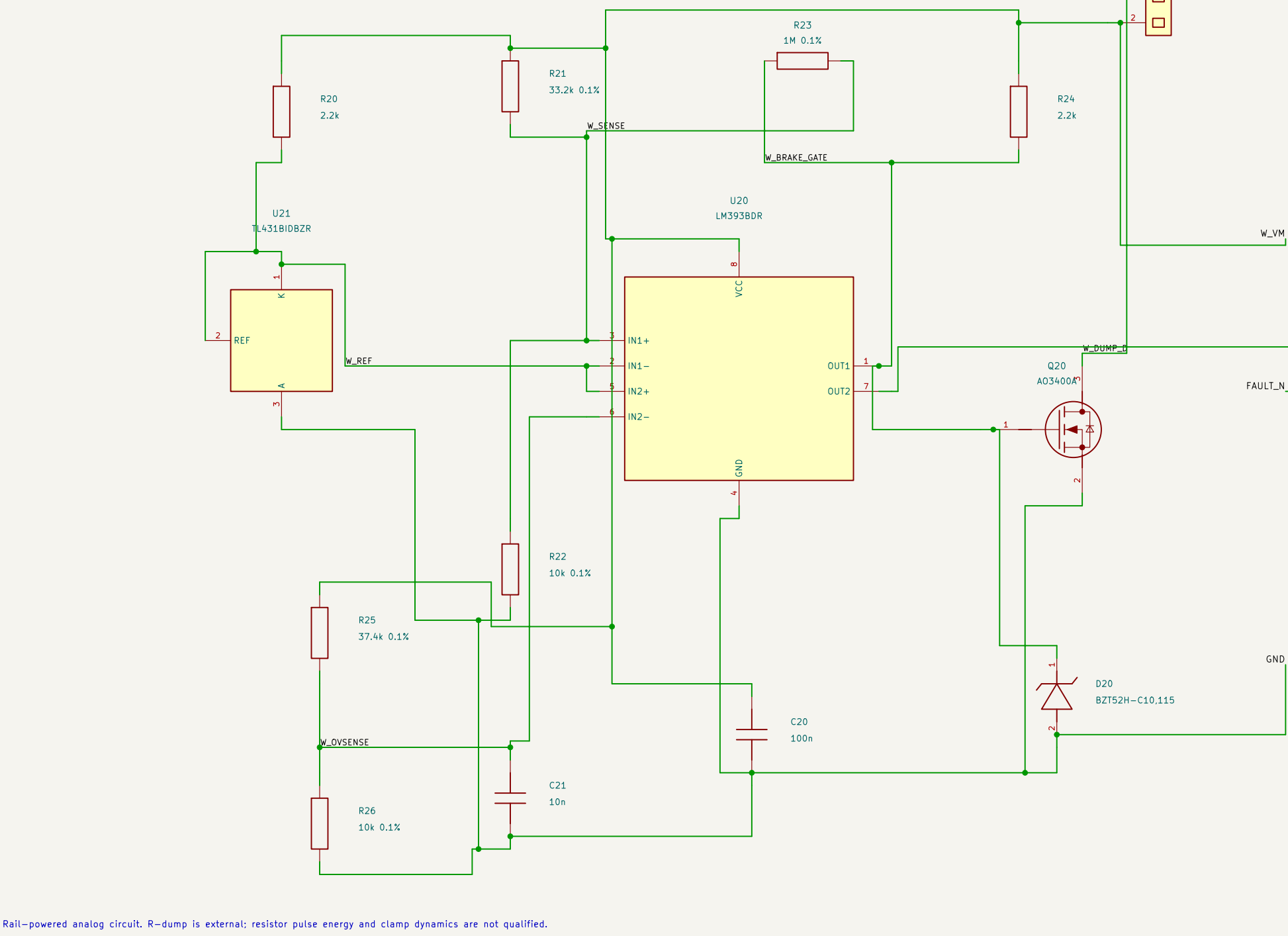


J2 -> 9V buck -> J3 | J4 -> 6V buck -> J5

J6 -> separate motion / interaction 5V converters

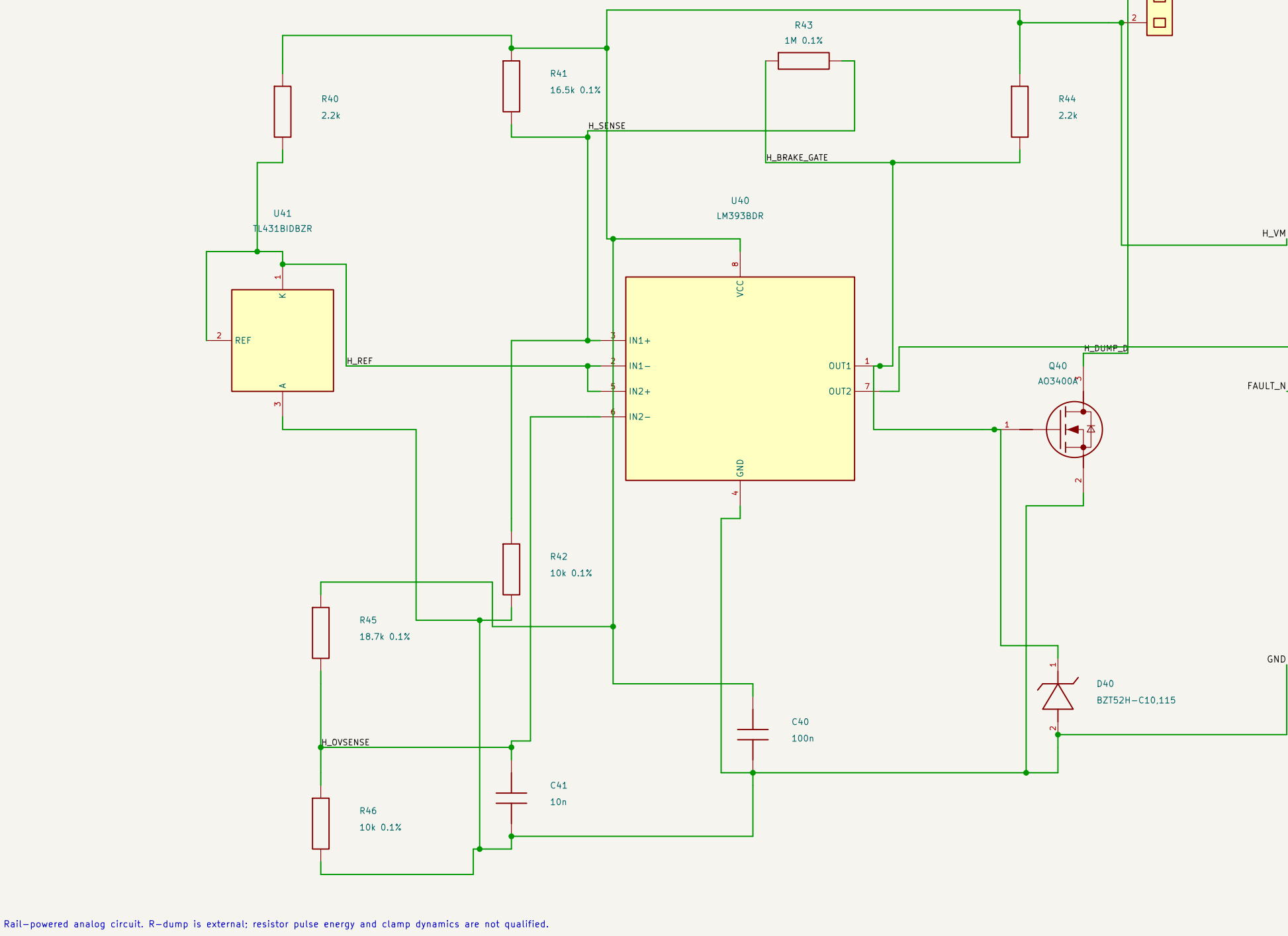
Connectors are boundaries only; buck converters, finished pack, 3S charger and USB–C negotiation are external.

05 Wheel regenerative clamp / overvoltage fault



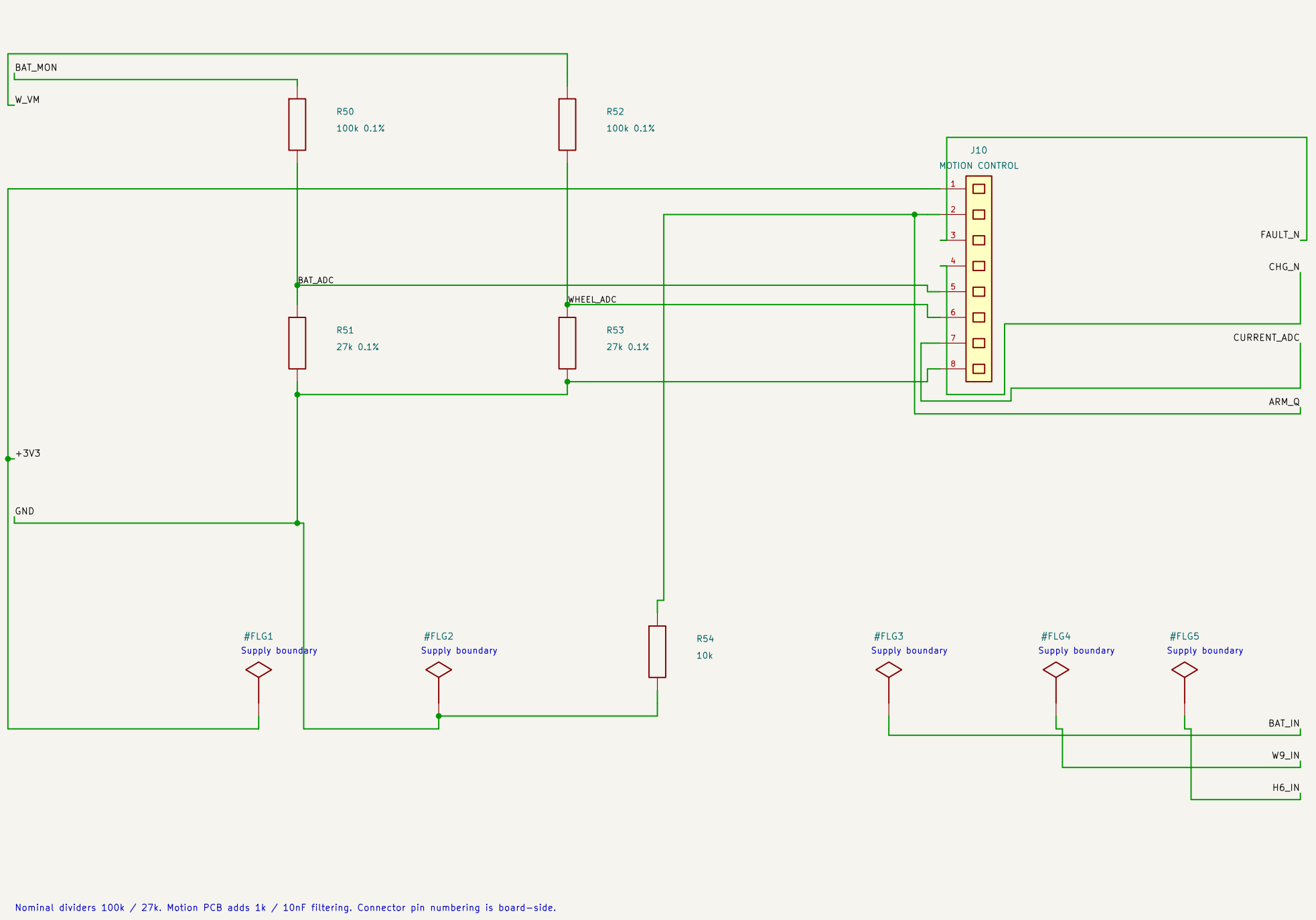
Rail–powered analog circuit. R–dump is external; resistor pulse energy and clamp dynamics are not qualified.

06 Head regenerative clamp / overvoltage fault



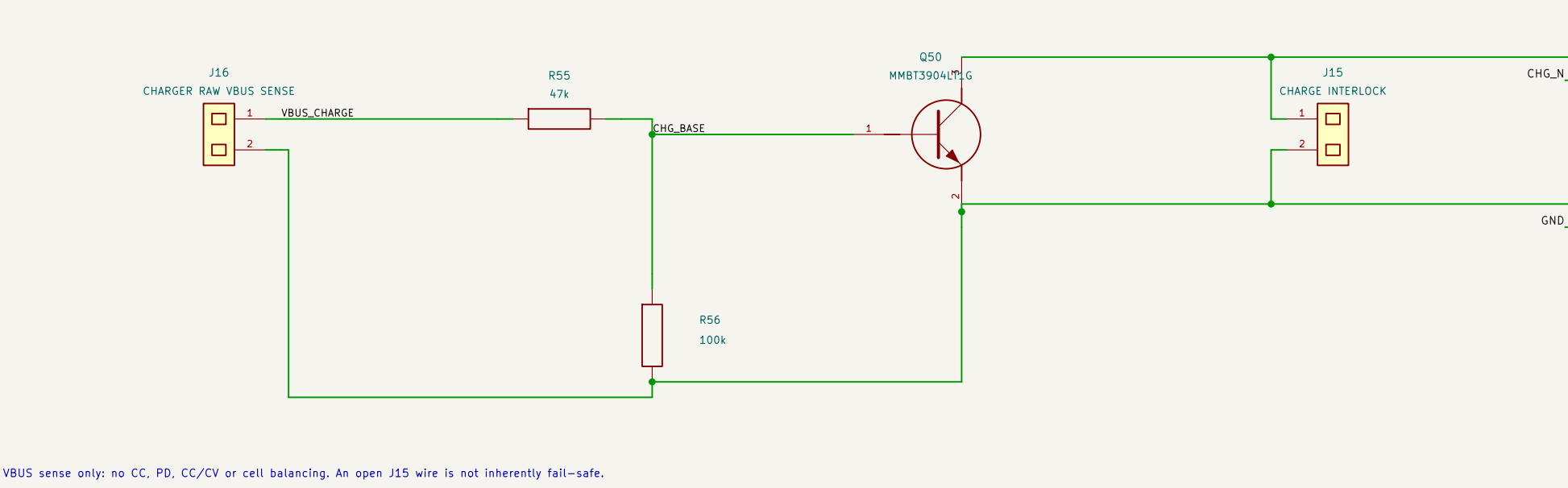
Rail–powered analog circuit. R–dump is external; resistor pulse energy and clamp dynamics are not qualified.

07 Voltage dividers / motion–board control connector



Nominal dividers 100k / 27k. Motion PCB adds 1k / 10nF filtering. Connector pin numbering is board–side.

08 Charger–present / insertion interlock



VBUS sense only; no CC, PD, CC/CV or cell balancing. An open J15 wire is not inherently fail–safe.

09 Mounting / release boundaries

